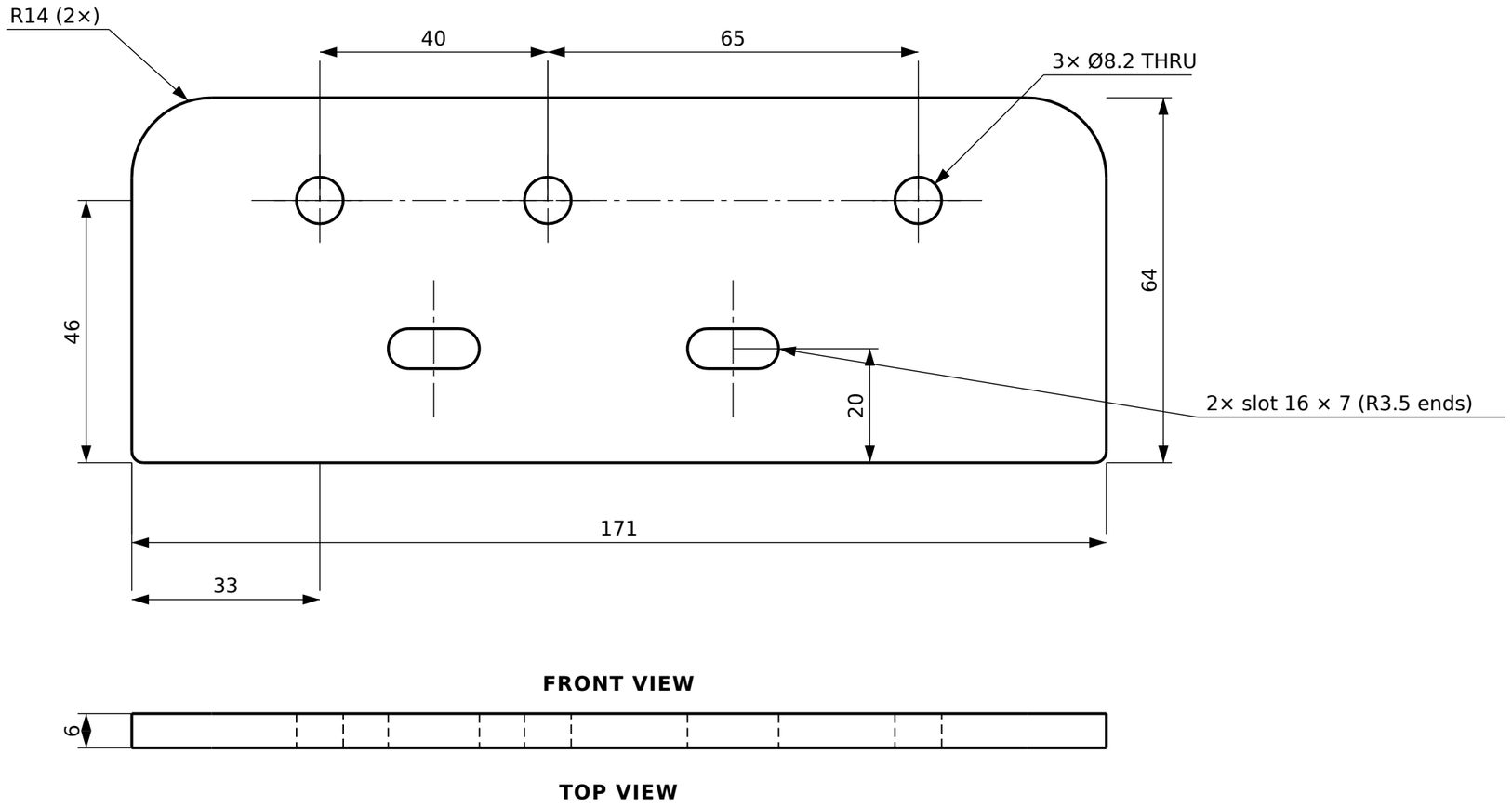


FRONT VIEW

TOP VIEW

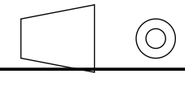


LEFT VIEW

Bore centres follow the gear centre distances a = m(z₁ + z₂)/2 = 40 and 65 mm (module 2; z = 15, 25, 40).

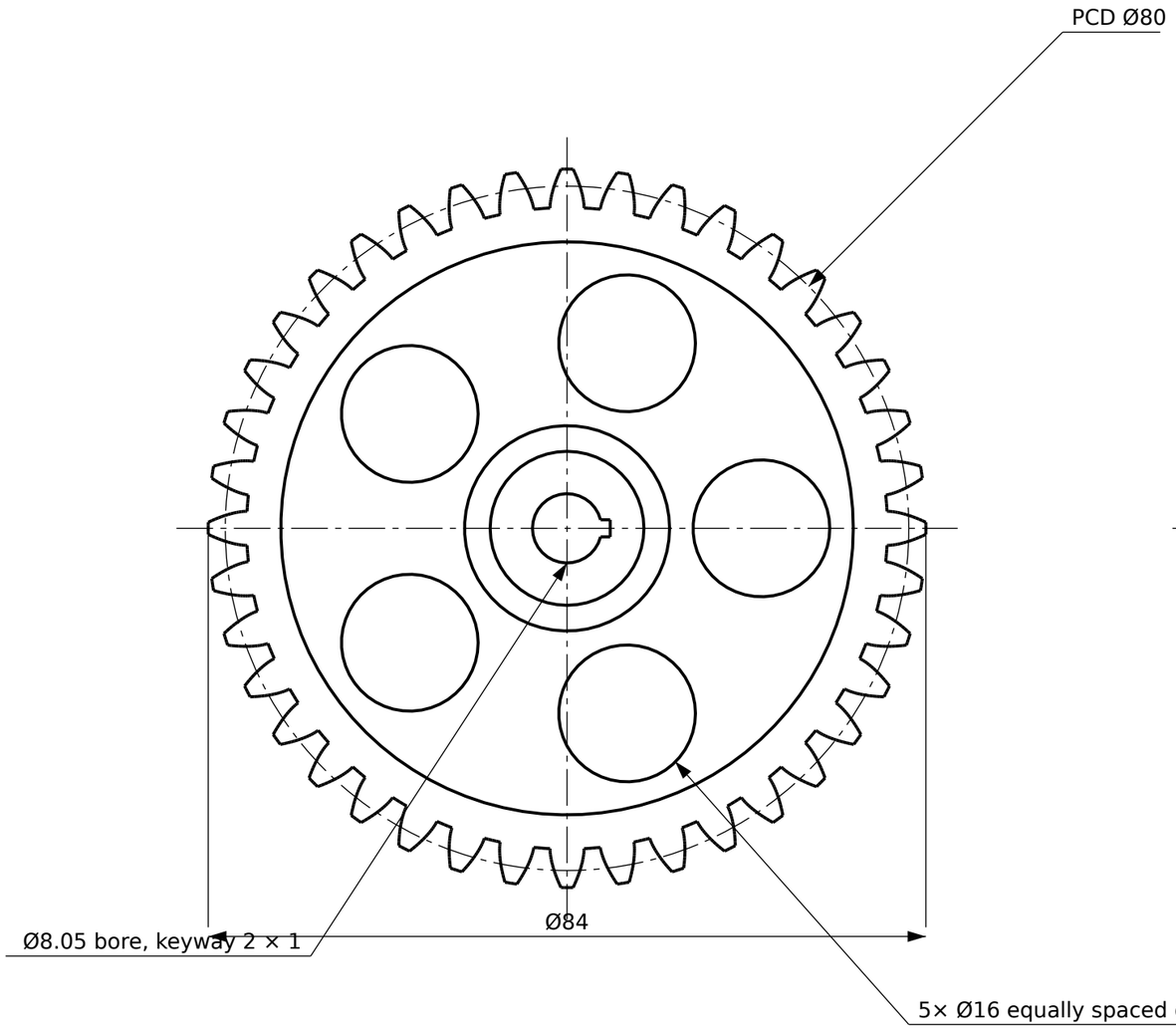
Bores Ø8.2 give 0.1 mm radial running clearance on the Ø8 shafts. Slot centres midway between bores. Break sharp edges 0.3 × 45°.

First-angle projection

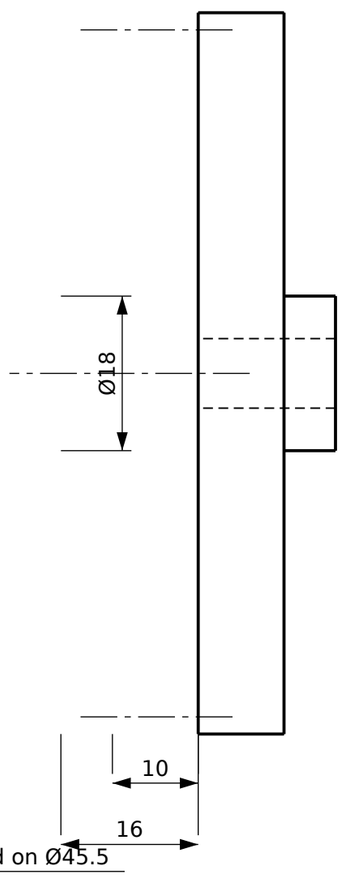


SIDE PLATE

Part no. GT-002	Scale 1:1
Material: Aluminium 6061-T6, 6 mm plate	Sheet 1 of 3 · A3
General tolerances ISO 2768-m · dimensions in mm	Drawn: Priyanshu Sharan · 2026-11-06



FACE VIEW (1.4 : 1)



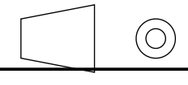
SIDE VIEW (1.4 : 1, conventional)

GEAR DATA	
Module m	2
Number of teeth z	40
Pressure angle	20°
Pitch circle Ø	80
Tip circle Ø	84
Root circle Ø	75
Face width	10
Backlash (tooth thickness)	0.1
Mates with	idler z = 25 at a = 65

Involute profile generated from first principles (src/parts.py); verified with zero overlap through a full mesh cycle (results/mesh-check.json).

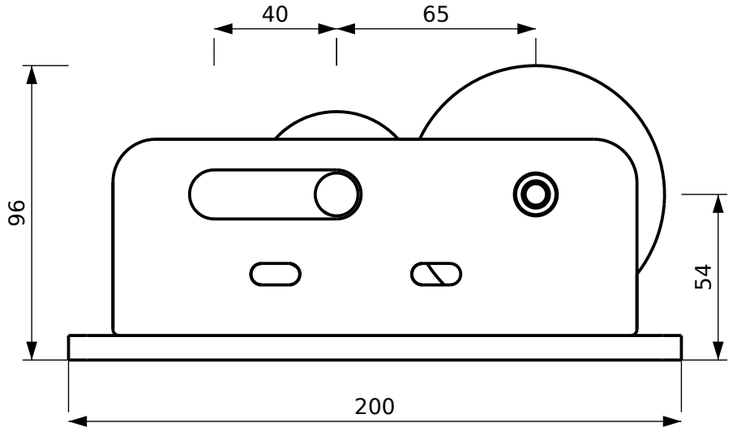
Web 3 mm thick between hub and rim (recess Ø67 × 7 deep from the hub side). Teeth: tip and root edges unbroken.

First-angle projection



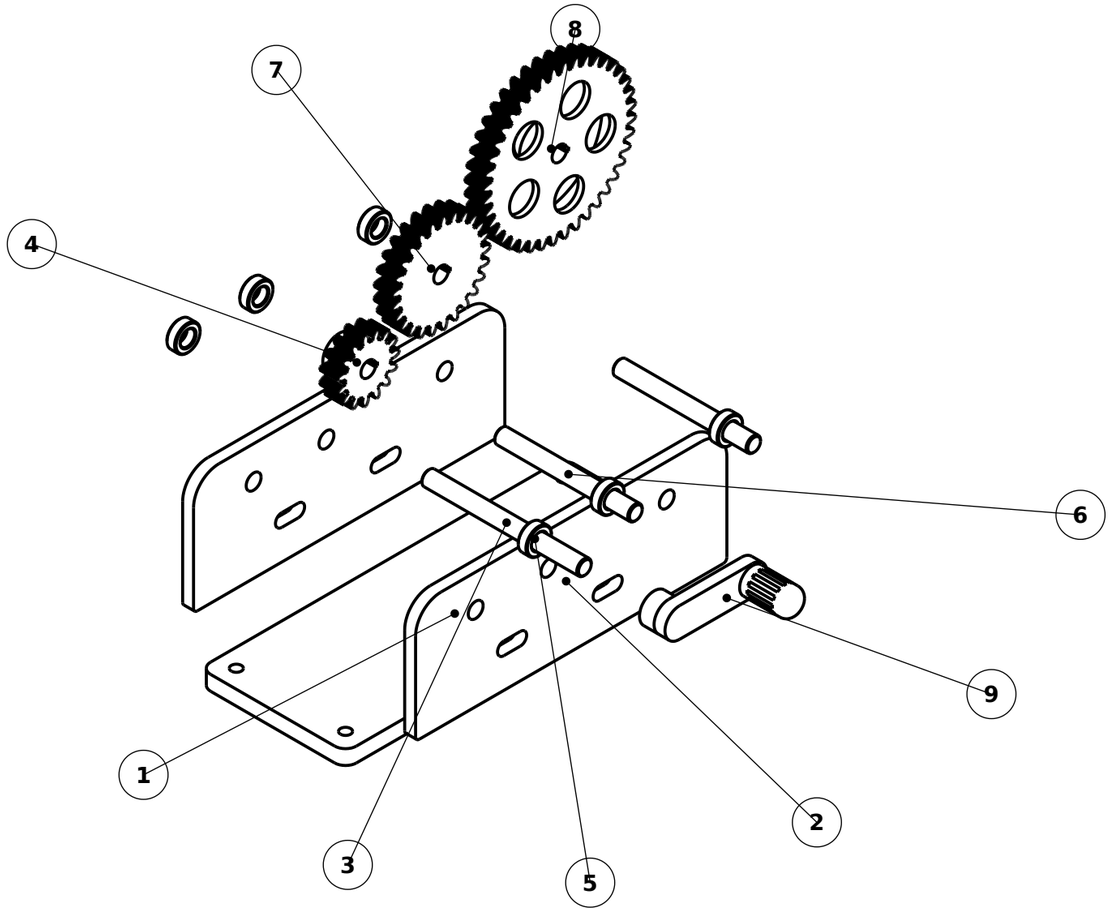
SPUR GEAR z = 40

Part no. GT-005	Scale 1.4:1
Material: Brass CW614N	Sheet 2 of 3 · A3
General tolerances ISO 2768-m · dimensions in mm	Drawn: Priyanshu Sharan · 2026-11-06



FRONT VIEW

TOP VIEW



EXPLODED ISOMETRIC VIEW

ITEM	PART	QTY	MATERIAL	MASS g
1	Base plate	1	aluminium anodised	342.9
2	Side plate	2	aluminium brushed	170.0
3	Input shaft Ø8 × 86	1	steel	33.9
4	Spur gear z15	1	brass	63.1
5	Shaft collar	6	steel black	3.9
6	Shaft Ø8 × 74	2	steel	29.2
7	Idler gear z25	1	acetal white	28.0
8	Spur gear z40	1	brass	219.2
9	Hand crank	1	aluminium red	26.1

First-angle projection

GEAR TRAIN ASSEMBLY

Part no. GT-000	Scale 1:2
Materials: see BOM	Sheet 3 of 3 · A3
General tolerances ISO 2768-m · dimensions in mm	Drawn: Priyanshu Sharan · 2026-11-06